

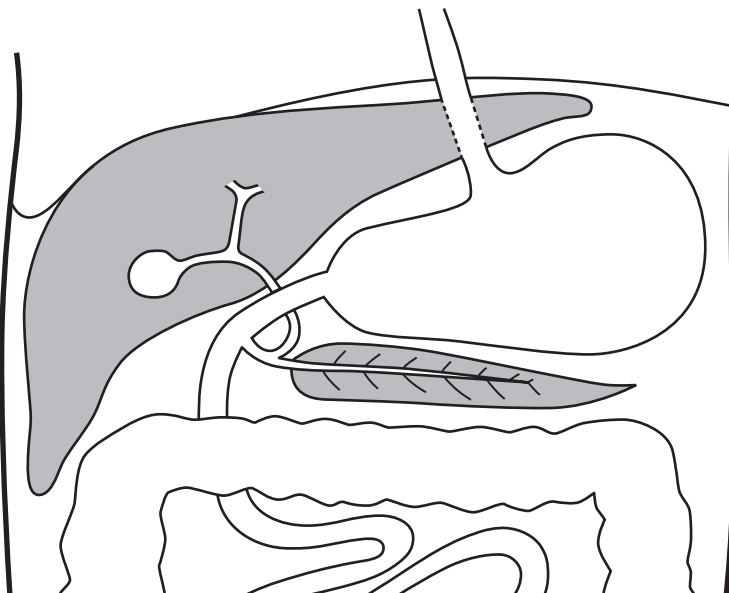
6. Cystic fibrosis is a genetic condition that affects many organs in the body including the pancreas. People with cystic fibrosis may have blockages in the pancreas which prevent enzymes from the pancreas reaching the small intestine. To overcome the problem of these blockages, people with cystic fibrosis take pancreatin tablets.

Pancreatin tablets have a tough coating that can resist low pH values. They contain a mixture of digestive enzymes normally produced in the pancreas.

- (a) State the name of the process that moves the tablet along the digestive system after it has been swallowed. [1]

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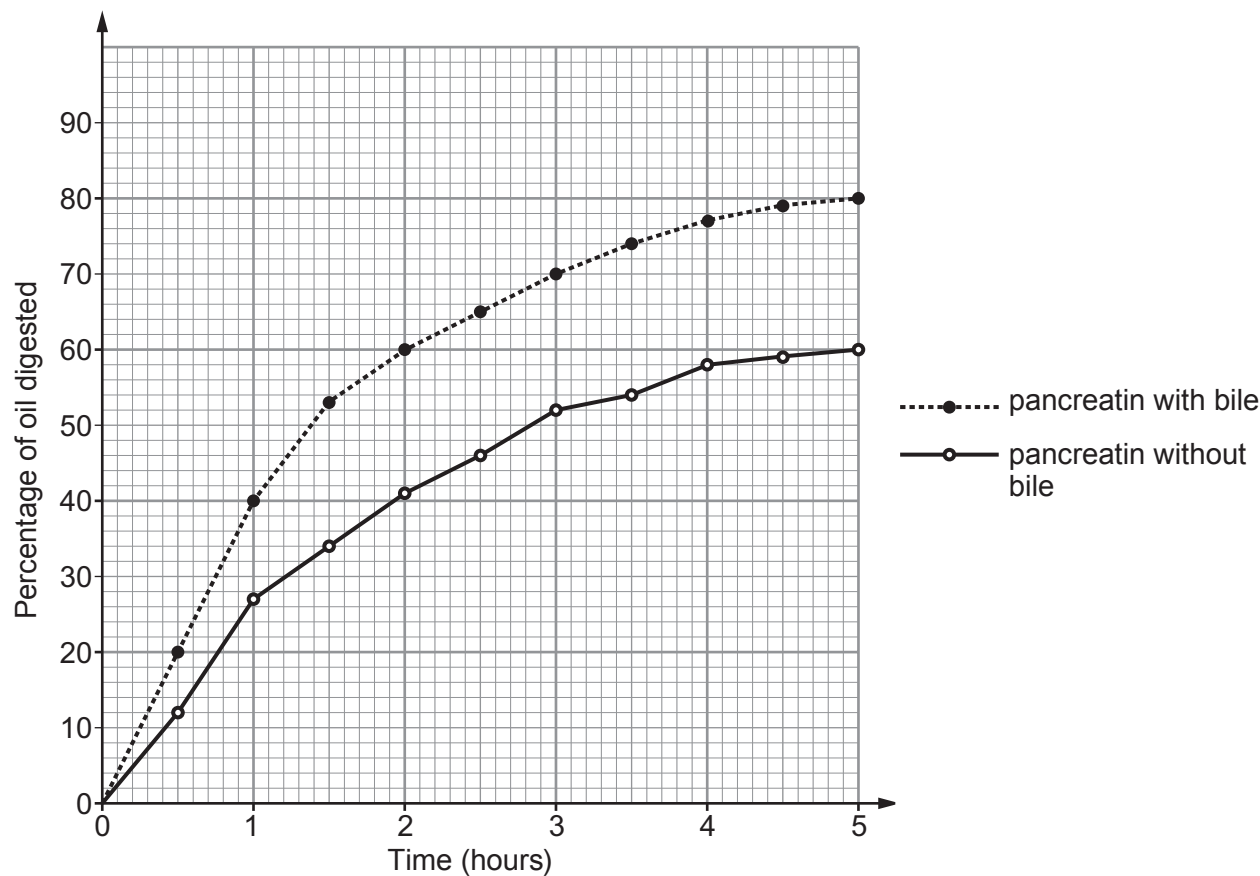
- (b) The diagram below shows part of the digestive system. [2]



- (i) Use a line labelled **X** to suggest the location of the blockage caused by cystic fibrosis.
- (ii) Use a line labelled **Y** to show the organ that secretes bile.



(c) The graphs below show the results of some experiments involving pancreatin.



Graph – Percentage of olive oil digested by pancreatin with and without bile.

(i) Pancreatin contains three different digestive enzymes, one of which is lipase. State the name of the *other two* enzymes present in pancreatin. [2]

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II.

(ii) Explain the effect of bile on the digestion of olive oil. [3]

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Examiner
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(iii) State the products formed when olive oil is digested and give **one** use of these products in the body. [2]

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(d) Suggest why pancreatin tablets need to have ‘a tough coating that can resist low pH values’. Explain your answer. [2]

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(e) State the name of the group of molecules to which enzymes belong and explain why the three enzymes in the pancreatin tablets have different structures. [2]

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